Excel Data Analyst Project Documentation

Retail Sales Data Analysis Using Excel

A Multi-Table Relational Analysis Built Entirely in Excel 2007

1. Project Overview

The objective of this project is to analyze retail sales data using Microsoft Excel 2007 and generate meaningful business insights through data cleaning, transformation, formula-based table merging, pivot analysis, and dashboard visualization.

Unlike a single flat dataset, this project works with **four relational tables** (Customer, Product, Store, and Sales) that had to be combined manually using lookup formulas, since Excel 2007 does not support Power Pivot or a native Data Model. This makes the project a strong demonstration of practical Excel skills under real version constraints.

This project demonstrates key Excel skills required for a Data Analyst role:

- Data Cleaning & Validation
- Relational Data Merging using VLOOKUP
- Formulas and Functions
- Pivot Tables & Pivot Charts
- Dashboard Creation (without Slicers / Timelines)
- Business Insights Generation

2. Business Problem

Management wants to understand the following business questions to drive data-driven decisions:

- Which product categories generate the highest revenue?
- Which regions and store types perform best?
- How do customer demographics and loyalty levels relate to spending?
- What are the yearly sales trends?
- Which payment methods are most commonly used by customers?

3. Dataset Information

Dataset Type: Relational Retail Sales Data | **Structure:** 4 Linked Tables | **Records:** 2,000+ Transactions

3.1 Customer Table

Column Name	Description
Customer_ID	Unique customer identifier
Name	Customer full name
Age	Customer age
Age_Group	Age category (Adult, Mid-age, Senior)
Gender	Customer gender
City	Customer city
State	Customer state
Country	Customer country
Loyalty_Level	Loyalty tier (Bronze, Silver, Gold, Platinum)

3.2 Product Table

Column Name	Description
Product_ID	Unique product identifier
Product_Name	Name of the product
Category	Product category (Electronics, Clothing, etc.)
Sub_Category	Product sub-category
Brand	Product brand / manufacturer
Cost	Cost price of the product
Stock Level	Current stock quantity and stock status

3.3 Store Table

Column Name	Description
Store_ID	Unique store identifier
Store_Name	Name of the store
Region	Sales region (East, West, North, South)
City	Store city
Store_Type	Store format (Flagship, Outlet, Online)

3.4 Sales Table (Transactions)

Column Name	Description
Sales_ID	Unique transaction identifier
Order_Date	Date of purchase
Customer_ID	Links to Customer table
Product_ID	Links to Product table
Store_ID	Links to Store table
Quantity	Units sold
Unit_Price	Price per unit
Discount	Discount applied on the order
Payment_Type	Mode of payment (COD, Credit Card, PayPal)
Total_Amount	Final transaction value after discount

3.5 Master Table (Merged Using VLOOKUP)

Since Excel 2007 does not support Power Pivot or relational Data Models, all dashboard charts and pivot tables were built from a single **Master Table**. This was created by using **VLOOKUP** to pull related fields from the Customer, Product, and Store tables directly into the Sales table — resulting in one flat, analysis-ready table with the following columns:

Sales_ID, Order_Date, Customer_ID, Product_ID, Store_ID, Age_Group, Gender, Product_Name, Category, Brand, Store, Region, Store_Type, Loyalty_Level, Quantity, Unit_Price, Discount, Payment_Type, Total_Amount

4. Data Cleaning Process

Step	Action
Remove Duplicates	Data tab -> Remove Duplicates, applied across all 4 source tables
Table Merging	VLOOKUP used to bring Customer, Product, and Store fields into the Sales table
Handle Missing Values	Blank cells identified and corrected before merging
Standardize Data	Inconsistent text values (e.g., region/category names) standardized
Date Formatting	Order_Date converted to proper Excel date format
Data Validation	Validation rules applied for Region, Category, Payment_Type fields

5. Excel Functions Used

Formula / Function	Purpose
=VLOOKUP()	Merging Customer, Product, and Store data into the Sales table
=SUM(Total_Amount)	Total Sales calculation
=AVERAGE(Total_Amount)	Average Order Value
=COUNTA(Sales_ID)	Total Orders count
=MAX() / =MIN()	Maximum and minimum sale value
=IF() / =IFS()	Logical categorization (e.g., stock level, age group)
=COUNTIF() / =SUMIF()	Conditional count and sum by category/region
=IFERROR()	Error handling on lookup formulas

6. Pivot Table Analysis

6.1 Sales by Region

Region	Revenue (₹)
North	6,43,579
South	6,34,369
East	6,23,106
West	2,70,614

***Insight:** North region generated the highest sales revenue, while West lags noticeably behind the other three regions.*

6.2 Sales by Category

Category	Revenue (₹)
Sports	5,42,010
Electronics	5,04,863
Home	4,24,795
Clothing	3,78,380
Beauty	3,21,619

***Insight:** Sports is the top-performing category, narrowly ahead of Electronics, together contributing nearly half of total revenue.*

6.3 Customer Segment Analysis (by Loyalty Level)

Loyalty Level	Revenue (₹)
Bronze	5,22,610
Silver	4,58,899
Gold	5,17,571
Platinum	6,72,587

***Insight:** Platinum loyalty customers contribute the highest revenue, suggesting loyalty tier correlates with spending.*

6.4 Store Type Comparison

Store Type	Revenue (₹)
Online	9,14,148
Flagship	8,47,118
Outlet	4,10,401

***Insight:** Online and Flagship stores significantly outperform Outlet stores in total revenue.*

7. Visualizations Created

Chart Type	Purpose	Key Insight
Line Chart	Yearly sales trend (2023-2025)	Sales rose sharply in 2024, then declined in 2025
Horizontal Bar Chart	Compare category-wise sales	Sports and Electronics lead all categories
Column Chart	Compare sales across regions	North region has the highest revenue
Donut Chart	Customer segment contribution	Platinum tier customers contribute the most
Column Chart	Store type comparison	Online and Flagship outperform Outlet stores
Pie Chart	Payment method share	Credit Card and PayPal are the most-used methods

8. Dashboard Components

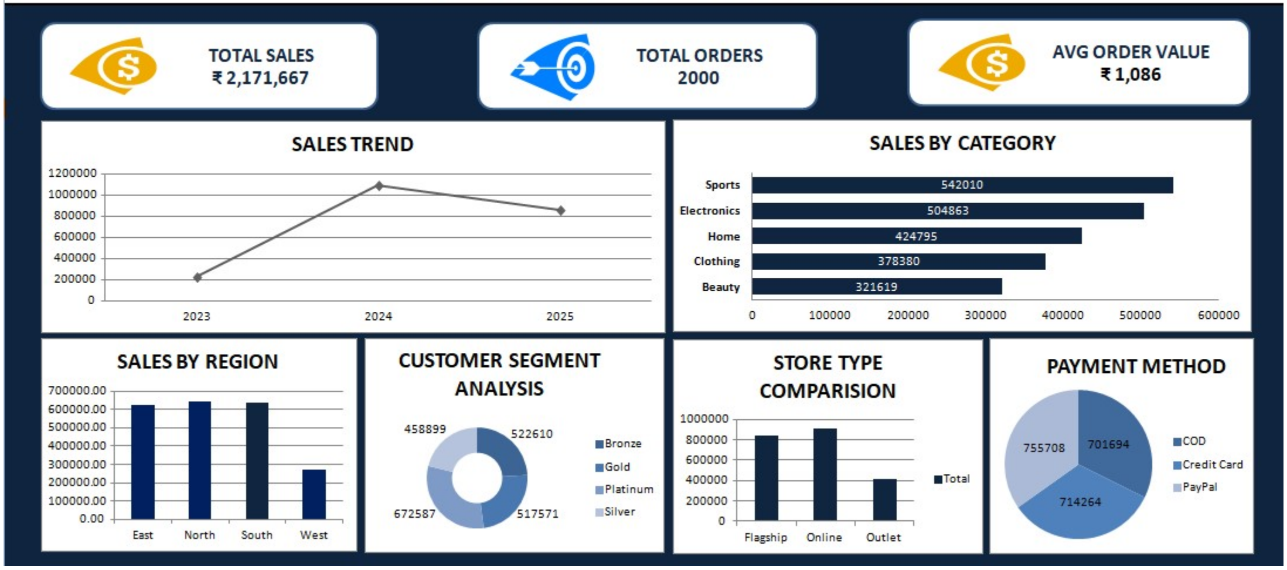
A sales dashboard was created using the components below. As this project was built in **Excel 2007**, Slicers and Timeline filters (introduced in later Excel versions) were not available, so the dashboard relies on KPI cards and pivot-based charts for a clear, single-view summary.

- **KPI Cards** — Total Sales, Total Orders, Average Order Value
- **Pivot Charts** — Linked to the Master Table
- **Sales Trend Chart** — Year-wise revenue movement
- **Category, Region, Store Type, Loyalty & Payment Method Breakdowns** — via dedicated pivot charts

KPI Summary

₹ 21,71,667	2,000	₹ 1,086
Total Sales	Total Orders	Avg Order Value

SALES DASHBOARD OVERVIEW



9. Key Business Insights

1. North region contributes the highest revenue among all four regions, with West trailing significantly behind.

2. Sports and Electronics are the two leading categories, together driving close to half of total business revenue.

3. Platinum and Bronze loyalty customers generate the most revenue, indicating that both high-value retained customers and a broad new-customer base matter to overall sales.

4. Sales grew sharply from 2023 to 2024 but declined in 2025, signaling a slowdown that warrants further investigation.

5. Online and Flagship store formats far outperform Outlet stores, suggesting a shift in customer preference away from physical outlet locations.

6. Credit Card and PayPal are the dominant payment methods, with COD used comparatively less.

10. Recommendations

Area	Recommendation
Sports & Electronics	Expand inventory and run targeted promotions on these top categories
West Region	Investigate underperformance and consider increased marketing investment
2025 Sales Decline	Review pricing, competition, and demand shifts driving the year-over-year drop
Outlet Stores	Reassess outlet store strategy given the gap versus Online and Flagship formats
Loyalty Program	Strengthen Silver/Gold tier engagement to push more customers toward Platinum-level spend
Payment Experience	Continue optimizing Credit Card / PayPal checkout flows given their high usage share

11. Skills Demonstrated

- ✓ Data Cleaning & Validation across multiple linked tables
- ✓ Relational Data Merging using VLOOKUP (Power Pivot alternative for Excel 2007)
- ✓ Excel Functions & Formulas
- ✓ Pivot Tables & Pivot Charts
- ✓ Dashboard Design under version constraints (no Slicers / Timelines)
- ✓ KPI Reporting
- ✓ Data Visualization
- ✓ Business Insights Generation

12. Conclusion

This Excel Data Analyst project successfully transformed four separate, relational retail datasets into a single, analysis-ready master table and a clear business dashboard — entirely within the constraints of Excel 2007. By combining VLOOKUP-based table merging with Pivot Tables, formulas, and chart-based visualizations, the project demonstrates that meaningful, decision-ready insights can be produced without relying on newer Excel features such as Power Pivot, Slicers, or Timelines. Management can use these insights to guide category investment, regional strategy, store-format planning, and loyalty program design.